SimCert: A Classical-Simulability Audit of Trained Quantum-NLP Text Classifiers

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Quantum natural language processing (QNLP) models, from compositional DisCoCat circuits to quantum self-attention networks and all-quantum token mixers, are promoted for a quantum benefit on text classification. That benefit is usually argued from accuracy, yet whether a *trained* QNLP circuit ever leaves the efficiently classically-simulable regime is rarely tested. The same question was recently settled for tabular quantum machine learning and for quantum convolutional networks, and the theory that trainable variational models are constrained matrix product states (MPS) makes it concrete for language too. We ask which published QNLP results survive a classical surrogate of the trained circuit.

We introduce SimCert, an open and model-agnostic protocol. It re-simulates each trained circuit as a bond-dimension-χ MPS, sweeps χ, and records the smallest value $\chi^\star$ at which a classical surrogate recovers the model's own predictions, together with an entanglement-removal ablation, the state entanglement entropy, a geometric-difference kernel test, and a data-reuploading spectral test. A single audit consumer treats every model uniformly through a common circuit representation, including a mixed-state purification path for models whose readout is a reduced density matrix.

We audit five published architectures, namely DisCoCat as realised in lambeq, the quantum self-attention models QSANN and QMSAN, the all-quantum token mixer CLAQS, and a variational reference classifier, across four datasets and up to ten qubits. A classical surrogate recovers the predictions at small and non-growing bond dimension. Attention and mixer models are recovered at χ=1, that is by a product state, and the compositional model at χ=2. As we grow the qubit count on a fixed task, $\chi^\star$ stays flat while the exact-MPS bound $2^{n/2}$ grows exponentially (Figure 1). The trained circuits carry entanglement, with mean bipartite entropy from about 0.1 to 1.3 nats, but removing it barely moves the result, so the entanglement is present without being load-bearing for the decision. The one revealing exception is the compositional model, whose grammar cups make χ=1 collapse toward chance, which is why it alone needs χ=2.

We read this cautiously. It suggests that the quantum resource is not the operative ingredient at the scales these models are tested, not that QNLP is classically simulable in general. We release the harness and phrase the finding as a protocol and a set of questions for QNLP model and benchmark design, chief among them which architectural choices, if any, make $\chi^\star$ grow with system size. A full write-up and the open-source harness accompany this abstract.

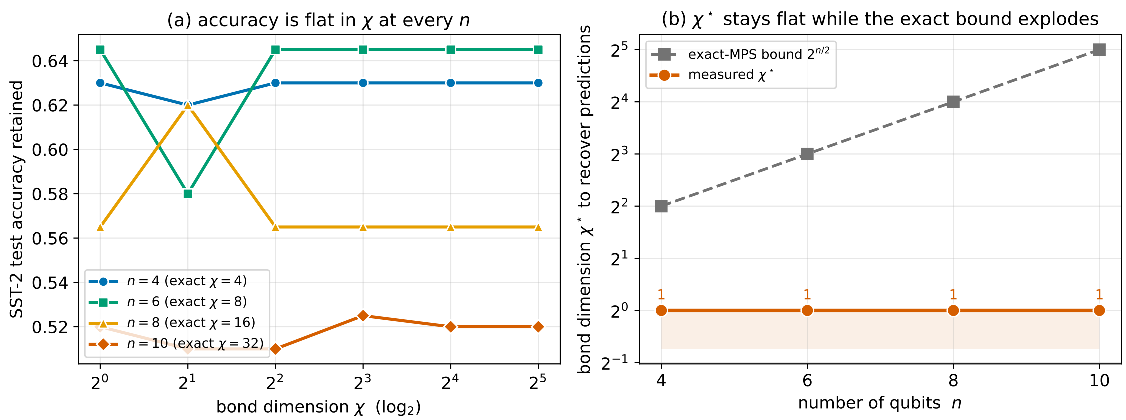

Figure 1: $\chi^\star$ stays flat at 1 as the reference model grows from four to ten qubits (right, orange), while the exact-MPS bound $2^{n/2}$ grows exponentially (grey). Left, accuracy is flat in χ at every size, so a product-state surrogate recovers the predictions throughout.